Acute Infectious diarrhoea

- Sometimes with vomiting, is the predominant symptom in infective gastroenteritis.
- The majority of episodes are due to infections spread by the faecal–oral route and transmitted either on fomites, on contaminated hands, or in food or water.
- Measures such as the provision of clean drinking water, appropriate disposal of human and animal sewage, and the application of simple principles of food hygiene can all limit gastroenteritis.
- The clinical features of food-borne gastroenteritis vary.
- Some organisms (Bacillus cereus, Staph. aureus and Vibrio cholerae) elute exotoxins that cause vomiting and/or so-called 'secretory' diarrhoea (watery diarrhoea without blood or faecal leucocytes, reflecting small bowel dysfunction).
- In general, the time from ingestion to the onset of symptoms is short and, other than dehydration, little systemic upset occurs
- Other organisms, such as Shigella spp., Campylobacter spp. and enterohaemorrhagic Escherichia coli (EHEC), may directly invade the mucosa of the small bowel or produce cytotoxins that cause mucosal ulceration, typically affecting the terminal small bowel and colon.
- The incubation period is longer and more systemic upset occurs, with prolonged bloody diarrhoea.
- Salmonella spp. are capable of invading enterocytes and of causing both a secretory response and invasive disease with systemic features.
- This is seen with Salmonella Typhi and Salmonella Paratyphi (enteric fever), but may occasionally be seen with other non-typhoidal Salmonellaspp., particularly in the immunocompromised host and the elderly.

13.9 Causes of infectious gastroenteritis

Toxin in food: <6 hrs incubation

- *Bacillus cereus*
- *Staph. aureus*
- *Clostridium* spp. enterotoxin

Bacterial: 12–72 hrs incubation

- Enterotoxigenic *Escherichia coli* (ETEC)
- Shiga toxin-producing *E. coli* (EHEC)*
- Entero-invasive *E. coli* (EIEC)*
- *Vibrio cholerae*
- *Salmonella*
- *Shigella**
- *Campylobacter**
- *Clostridioides difficile**

Viral: short incubation

- Rotavirus
- Norovirus

Protozoal: long incubation

- Giardiasis
- Cryptosporidiosis
- Microsporidiosis
- Amoebic dysentery*
- Cystoisosporiasis

*Associated with bloody diarrhoea.

13.10 Differential diagnosis of acute diarrhoea and vomiting

Infectious causes

- Gastroenteritis
- *Clostridioides difficile* infection
- Acute diverticulitis (p. 832)
- Sepsis (p. 198)
- Pelvic inflammatory disease (p. 375)
- Meningococcaemia
- Pneumonia (especially 'atypical disease', p. 512)
- Malaria

Non-infectious causes

Gastrointestinal

- Inflammatory bowel disease (p. 835)
- Bowel malignancy (p. 826)
- Overflow from constipation (p. 833)
- Enteral tube feeding

Metabolic

- Diabetic ketoacidosis (p. 721)
- Thyrotoxicosis (p. 652)
- Uraemia (p. 588)
- Neuro-endocrine tumours releasing (e.g.) VIP or 5-HT

Drugs and toxins

- NSAIDs
- Cytotoxic agents
- Antibiotics
- Proton pump inhibitors
- Dinoflagellates (p. 236)
- Plant toxins (p. 237)
- Heavy metals
- Ciguatera fish poisoning (p. 236)
- Scombrotoxic fish poisoning (p. 237)

(5-HT = 5-hydroxytryptamine, serotonin; NSAIDs = non-steroidal anti-inflammatory drugs; VIP = vasoactive intestinal peptide)

Clinical assessment

- The history should address foods ingested , duration and frequency of diarrhoea, presence of blood or steatorrhoea, abdominal pain and tenesmus, and whether other people have been affected.
- Fever and bloody diarrhoea suggest an invasive, colitic, dysenteric process.
- An incubation period of less than 18 hours suggests toxin-mediated food poisoning, and longer than 5 days suggests diarrhoea caused by protozoa or helminths.
- Person-to-person spread suggests certain infections, such as shigellosis or cholera. Examination includes assessment of the degree of dehydration.
- Assessment for early signs of hypotension, such as thirst, headache, altered skin turgor, dry mucous membranes and postural hypotension, is important, particularly in tropical regions where dehydration progresses rapidly.
- Signs of more marked dehydration include supine hypotension and tachycardia, decreased urinary output, delirium and sunken eyes.
- The blood pressure, pulse rate, urine output and ongoing stool losses should be monitored closely.

13.12 Foods associated with infectious illness, including gastroenteritis

Raw seafood	
• Norovirus • *Vibrio* spp.	• Hepatitis A
Raw eggs	
• *Salmonella* serovars	
Under-cooked meat or poultry	
• *Salmonella* serovars • *Campylobacter* spp. • EHEC	• Hepatitis E (pork products) • *Clostridium perfringens*
Unpasteurised milk or juice	
• *Salmonella* serovars • *Campylobacter* spp.	• EHEC • *Yersinia enterocolitica*
Unpasteurised soft cheeses	
• *Salmonella* serovars • *Campylobacter* spp. • ETEC	• *Yersinia enterocolitica* • *Listeria monocytogenes*
Home-made canned goods	
• *Clostridium botulinum*	
Raw hot dogs, pâté	
• *Listeria monocytogenes*	

(EHEC = enterohaemorrhagic *Escherichia coli*; ETEC = enterotoxigenic *E. coli*)

Investigations

- These include stool inspection for blood and microscopy for leucocytes, and also an examination for ova, cysts and parasites if the history indicates residence or travel to areas where these infections are prevalent. Stool culture should be performed and C. difficile toxin sought.
- FBC and serum electrolytes indicate the degree of inflammation and dehydration. Where cholera is prevalent, examination of a wet film with dark-field microscopy for darting motility may provide a diagnosis. In a malarious area, a blood film for malaria parasites should be obtained.
- Blood and urine cultures and a chest X-ray may identify alternative sites of infection, particularly if the clinical features suggest a syndrome other than gastroenteritis. Fluid replacement
- Replacement of fluid losses in diarrhoeal illness is crucial and may be life-saving. Although normal daily fluid intake in an adult is only 1–2 L, there is considerable additional fluid movement in and out of the gut in secretions .
- Altered gut resorption with diarrhoea can result in substantial fluid loss; for example, 10 20 L of fluid may be lost in 24 hours in cholera.
- The fluid lost in diarrhoea is isotonic, so both water and electrolytes need to be replaced.
- Absorption of electrolytes from the gut is an active process requiring energy. Infected mucosa is capable of very rapid fluid and electrolyte transport if carbohydrate is available as an energy source.
- Oral rehydration solutions (ORS) therefore contain sugars, as well as water and electrolytes .
- ORS can be just as effective as intravenous replacement fluid, even in the management of cholera. In mild to moderate gastroenteritis, adults should be encouraged to drink fluids and, if possible, continue normal dietary food intake.
- If this is impossible – due to vomiting, for example – intravenous fluid administration will be required.

i 13.13 Composition of oral rehydration solution and other replacement fluids*

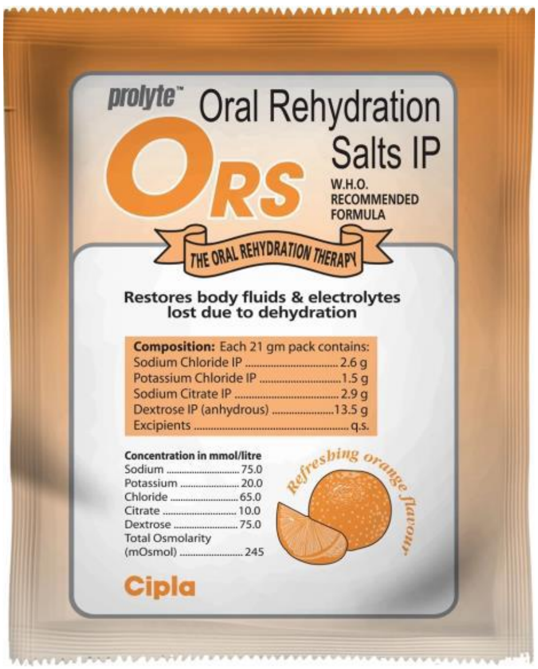
Fluid	Na	K	Cl	Energy
WHO	90	20	80	54
Dioralyte	60	20	60	71
Pepsi	6.5	0.8	–	400
7UP	7.5	0.2	–	320
Apple juice	0.4	26	–	480
Orange juice	0.2	49	–	400
Breast milk	22	36	28	670

*Values given in mmol/L for electrolyte and kcal/L for energy components.
(WHO = World Health Organization)

- In very sick patients or those with cardiac or renal disease, monitoring of urine output and central venous pressure may be necessary.
- The volume of fluid replacement required should be estimated based on the following considerations:

Intravenous Crystalloid Solutions

	Na^+	K^+	Ca^{2+}	Mg^{2+}	Cl^-	Osmolarity
Human Plasma	135-145	4.0-5.0	2.2-2.6	1.0-2.0	95-110	291
0.9% normal saline	154	0	0	0	154	308
Lactated ringer's	130	4.5	2.7	0	109	280



***Replacement of established deficit*.**

- After 48 hours of moderate diarrhoea (6–10 stools per 24 hrs), the average adult will be 2–4 L depleted from diarrhoea alone.
- Associated vomiting will compound this.
- Adults with this symptomatology should therefore be given rapid replacement of 1–1.5 L, either orally (ORS) or by intravenous infusion (normal saline), within the first 2–4 hours of presentation.
- Longer symptomatology or more persistent/severe diarrhoea rapidly produces fluid losses comparable to diabetic ketoacidosis and is a metabolic emergency requiring active intervention.

Replacement of ongoing losses.

- The average adult's diarrhoeal stool accounts for a loss of 200 mL of isotonic fluid.
- Stool losses should be carefully charted and an estimate of ongoing replacement fluid calculated.
- Commercially available rehydration sachets are conveniently produced to provide 200 mL of ORS; one sachet per diarrhoea stool is an appropriate estimate of supplementary replacement requirements.

Replacement of normal daily requirement.

- The average adult has a daily requirement of 1–1.5 L of fluid in addition to the calculations above.
- This will be increased substantially in fever or a hot environment.

Antimicrobial agents

- In non-specific gastroenteritis, routine use of antimicrobials does not improve outcome and may lead to antimicrobial resistance or side-effects.
- They are usually used where there is systemic involvement, a host with immunocompromise or significant comorbidity.
- Evidence suggests that, in EHEC infections, the use of antibiotics may make the complication of haemolytic uraemic syndrome more likely due to increased toxin release.
- Antibiotics should therefore not be used in this condition.
- ***Conversely, antibiotics are indicated in Shigella dysenteriae infection and in invasive salmonellosis – in particular, typhoid fever.***
- ***Antibiotics may also be advantageous in cholera epidemics, reducing infectivity and controlling the spread of infection.***

Antidiarrhoeal, antimotility and antisecretory agents

- These agents are not usually recommended in acute infective diarrhoea. Loperamide, diphenoxylate and opiates are potentially dangerous in dysentery in childhood, causing intussusception.
- Antisecretory agents, such as bismuth and chlorpromazine, may make the stools appear more bulky but do not reduce stool fluid losses and may cause significant sedation. Adsorbents, such as kaolin or charcoal, have little effect.

" لا تنسوا الدعاء لأهالينا في فلسطين "